Literate Programming with Quarto

Mahdi Abu Atmeh (401054029)

abu_atmeh.mahdi@stud.hs-fresenius.de

Fresenius University of Applied Science

Github.com/MadAtmeh/LPQ

Literate Programming

Is a programming approach that combines code and explanation in a single document. It prioritizes programs to be written for **humans** *first* then **machines** *second*.

Motivation:

Data analysis ≠ just code

Quarto

Is an open-source publishing system that supports literate programming by combining **code**, **text**, and **output** in a single document. It allows users to create:

History: Literate Programming

It was Introduced in the early 1980s by **Donald Knuth**

- Computer scientist and author of *The Art of Computer Programming*.
- Developed the concept while writing the book *The TeXbook* and building the **TeX system**.
- Formalized in his work: Literate Programming (1984) supported alongside tools like **WEB** (for Pascal) and **CWEB** (for C).
- Advocated that programming is a form of communication and storytelling.

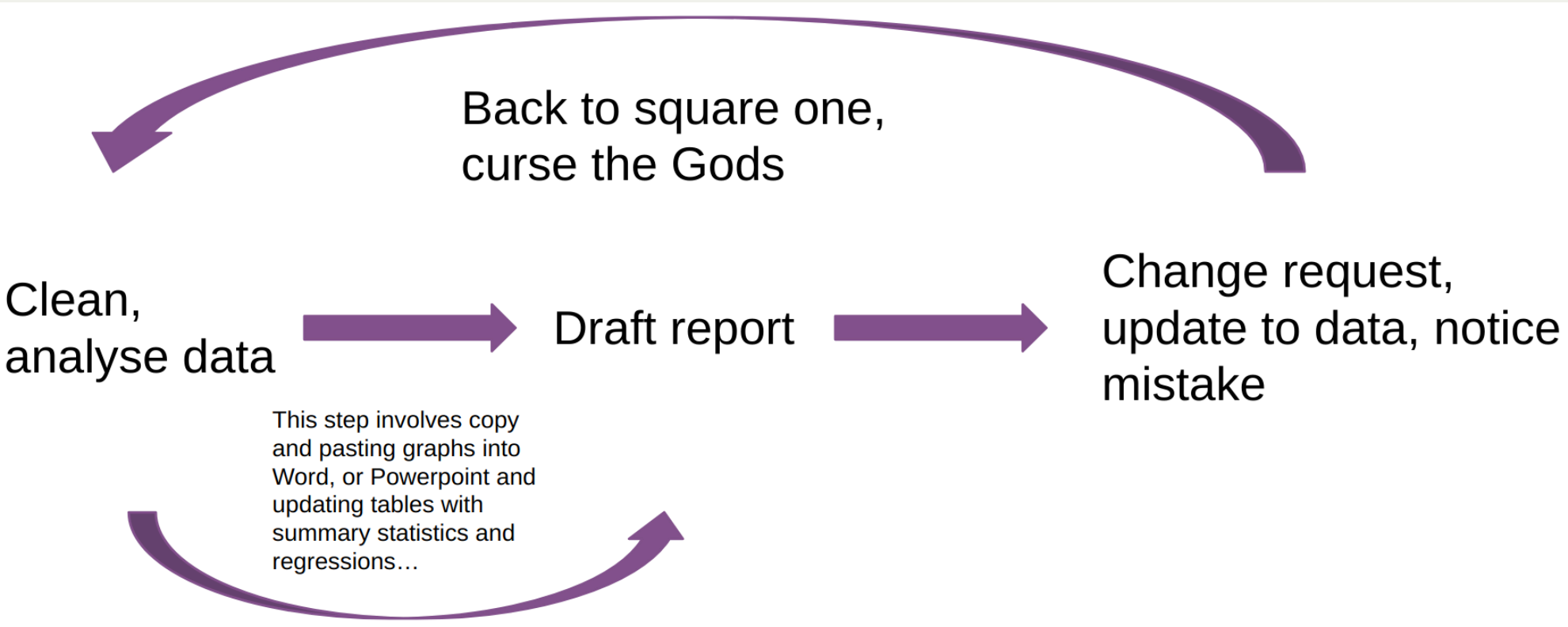
Why Literate Programming Matters

- Improves **reproducibility** (results can always be regenerated)
- Reduces **errors** from copying code or manual steps
- Makes analysis **easier to understand and share**
- Combines **thinking + coding + reporting**
- Saves time in long-term projects

Where is Literate Programming Used?

- Data science reports
- Academic research papers
- Business dashboards & analytics
- Scientific experiments
- Coursework & assignments

The Problem



This leads to **repetitive work, high error risk, poor reproducibility and extremely time-consuming**.

Core Concept: Literate Programming

Literate programming is the idea of combining:

- **code** (what you run)
- **data** (what you analyze)
- **output/analysis** (results like tables or plots)
- **documentation** (explanations in plain language)

= All in one single document

Example: Understanding Literate Programming

```
# Documentation:
# This example analyzes student scores to understand overall performance.

# Data:
scores <- c(70, 75, 80, 90, 95)

# Code:
average <- mean(scores)

# Output:
average

# Analysis:
# The average score is 82, indicating that students performed well overall.
```

Historical Tools & Evolution

Early tool: **Sweave**

It combines **R** and **LaTeX**, allowing users to embed code directly within documents for reproducible research.

Evolution: **Knitr**

Sweave later evolved into knitr, which improved *usability*, *flexibility*, and overall *performance*. This evolution enabled:

- Easier customization
- Better integration of code, results and narrative text

Modern Tools

Quarto

The latest in literate programming for **R** is a new tool developed by Posit. It is designed to make creating dynamic documents easier and more organized. It supports:

- R
- Python
- Julia

Using Quarto with R

Step 1: Install required software:

1. **R** – A programming language for data analysis
2. **RStudio** – An easy-to-use interface for writing R code and working with documents
3. **Quarto** – A tool to create documents that combine text, code, and results
4. **Git** (optional) – Helps track changes and manage versions of your work

Step 2: Install required R Packages:

- **tidyverse** - for data analysis install.packages("tidyverse")
- **knitr** - for rendering install.packages("knitr")
- **rmarkdown** - for markdown setup install.packages("rmarkdown")
- **tinytex** - for PDF support install.packages("tinytex") + tinytex:install_tinytex()

Running these commands in R

Simply run these commands in an R console to get started:

```
install.packages("tidyverse")
install.packages("knitr")
install.packages("rmarkdown")
install.packages("tinytex")

# Install TinyTeX (for PDF rendering)
tinytex:install_tinytex()

# Load libraries
library(tidyverse)
library(knitr)
library(rmarkdown)
library(tinytex)
```

Markdown Ultrabasics

Feature	Syntax Example	Description
Heading 1	`# Heading`	Main title
Heading 2	`## Heading`	Section
Heading 3	`### Heading`	Subsection
Italic	`*text*` or `_text_`	Italic text
Bold	`**text**` or `__text__`	Bold text
Strikethrough	`~~text~~`	Deleted text

Quarto File Structure (.qmd)

A **Quarto** document (.qmd) is made up of:

- **YAML header** → metadata (title, format, author)
- **Markdown text** → written explanation
- **Code chunks** → executable analysis

Overall, it combines **text**, **code**, and **results** in one file.

Code Chunks

Are sections where executable code is written inside a document. This is the core feature of literate programming. They allow:

- Code execution inside the document
- Reproducible results
- Combined explanation + output

Code Chunk Options

You can control how code appears using options:

Option	Effect
`echo: false`	Hide code
`eval: false`	Do not run code
`warning: false`	Hide warnings
`message: false`	Hide messages

Purpose: Code Chunk Options

These options are used to:

- Improve readability
- Clean up output
- Focus on results instead of code
- Customize document appearance

Example: Code Chunks I

1. Content Chunk

```
## What is Literate Programming?
- Programming is a form of communication
- Code and explanation are written together
```

2. Code Chunk

```
scores <- c(70, 80, 90)
mean(scores)
```

3. Analysis chunk

```
## Average Score Example

scores <- c(70, 75, 80, 90, 95)
mean(scores)
```

Example: Code Chunks II

4. Image Chunk

```
![](Image name.png){fig-align="Position"}
```

5. Speaker Notes

```
::: notes
Emphasize that the goal is readability, not just execution.
:::
```

Data Analysis Workflow in R

This is the standard workflow in R for data analysis.

Import Data → load data using `read_csv()`

Clean Data → use dplyr to filter and transform data

Analyze Data → apply statistical methods

Visualize Data → create plots using ggplot2

Full Code Demo

```
# ------------------------------
# Documentation:
# This example analyzes student exam scores to understand class performance.
# We compute summary statistics and visualize the distribution.

# ------------------------------
# Data:
scores <- c(55, 62, 70, 75, 80, 85, 90, 95, 100)

# ------------------------------
# Code:
mean_score <- mean(scores)
median_score <- median(scores)
sd_score <- sd(scores)
```

Exercise

- **Task 1:** Use the same structure from the demonstration code to analyze a new set of student scores.

New Dataset: scores <- c(48, 52, 67, 73, 78, 81, 88, 92, 96)

- **Task 2:** Visualize the data.

Thank you for Listening!